

Arsh Dalal

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EDUCATION

The University of Connecticut | Storrs, CT

Bachelor of Science in Computer Science | GPA: 3.5/4.0

May 2027

Relevant Coursework: DSA, AI/ML, Systems Programming, Computer Architecture, Computer Networking, Cloud Computing

EXPERIENCE

Otis Elevator Co. | Farmington, CT

May 2025 - Present

IoT Intern

- Built and deployed a full-stack platform using ASP.NET, React, PostgreSQL, and Azure to track data across thousands of elevator systems, reducing manual data retrieval by 10%
- Developed Python and .NET Core components for a cloud-connected elevator automation platform supporting thousands of elevators, helping field mechanics remotely access controller data and diagnostic workflows.
- Implemented distributed messaging workflows with Azure Service Bus and Redis, reducing diagnostic data retrieval time from ~15 minutes to ~2 minutes while coordinating script execution events across cloud services.
- Developed a sales automation system (Power Automate + Virtual Agents) reducing order processing time by 20%.

ProPark Mobility | Hartford, CT

Jan 2024 - Apr 2025

Event Coordinator

- Managed logistics and operations for events with 20,000+ attendees, ensuring reliability under high-load conditions.
- Enhanced team efficiency by 20% by implementing structured performance feedback systems and standardized event debrief templates across all teams.

TECHNICAL PROJECTS

BGP Internet Routing Simulator | C++, Systems, Networking, Multithreading

Spring 2026

- Built a scalable BGP simulator modeling 80K+ nodes and ~300K edges in ~2 seconds, implementing real-world routing policies.
- Engineered low-level optimizations including AVX2-accelerated SIMD for path-loop detection and batched mutex locking, reducing lock contention by 10x.
- Designed a custom memory-efficient architecture using integer prefix mapping and pre-allocated queues to minimize overhead during large graph traversals.

Google Docs Clone | Python, Flask, Socket.IO, MySQL, React, Redis

Fall 2025

- Built a real-time collaborative editor supporting multi-user synchronization, live autosave, and custom encryption while utilizing Redis to optimize concurrent user data handling.
- Designed scalable backend handling concurrent connections via WebSockets with a secure registration system.

UConn Personalized Meal Planning | Python, JS, OpenAI API, Web Scraping

Fall 2024

- Built a web app for UConn students to create personalized meal plans based on dietary needs, pulling live data from dining hall menus to generate goal-aligned food schedules.

ACTIVITIES

UConn AI and Machine Learning Club

Spring 2025 - Present

- Built an AI-powered clothing detection system that analyzes images to identify apparel and find similar products online retailers.
- Applied generative AI-assisted development practices to accelerate implementation, debugging, and code iteration while managing application quality and maintainability.

SKILLS

- **Programming Languages:** Python, C++, C, C#, JavaScript, SQL, R, HTML/CSS
- **Frameworks and Libraries:** React, Flask, Django, ASP.NET, ADO, .NET Core, SocketIO,
- **Cloud and Technologies:** Azure DevOps, Git/GitHub, Claude Code/Copilot, Docker, AWS, Redis, Wireshark